4. Deficiency of zinc may interfere with storage and secretion of insulin.

5. Zinc deficiency causes alopecia (hair loss)



*Figure 9.19 alopecia*

### 9.3.6.Cobalt

Cobalt is a component of vitamin B12. Elemental cobalt of the diet can be converted to Cobalamine by the intestinal bacteria. Cobalt occurs in small amount in all tissues, higher concentrations occur in liver and kidneys.

### Functions

Most of the cobalt is present in vitamin B12, which is necessary for maturation of red blood cells.

Cobalt is an activator of the enzyme Phosphoglucomutase and glycyl - glycine peptidase.

### Sources

It is largely available in food. Normal average diet contains about 5 to 8 micro grams of cobalt. One µg of vitamin B12 contains ~0.0225- 0.045 µg of cobalt.

### Absorption and excretion

Cobalt is absorbed from the small intestine. About 65% of ingested cobalt is excreted in the urine and the remaining in the feces.

### Deficiency

Cobalt deficiency is rare in human beings.

### 9.3.7. Manganese

Manganese is an essential trace element. It is present as $Mn^{2+}$ ion and found in high concentrations in the mitochondria.

### Functions

1. Manganese acts as a cofactor for number of enzymes including arginase, hexokinase and isocitrate dehydrogenase.
2. Manganese is essential for normal bone structure, reproduction and functioning of the central nervous system.
3. Manganese also functions with vitamin K in the formation of prothrombin.
4. It inhibits lipid peroxidation.
5. Manganese also participates in glycoprotein and proteoglycan synthesis.
6. It helps in Porphyrin synthesis.
7. Manganese is involved in fatty acid and cholesterol synthesis.



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